Steps to deploy React+Vite app to github

Step 1: Create a React + Vite project
If you already have a project, skip this step.

```
>>> npm create vite@latest my-app
>>> cd my-app
>>> npm install
>>> npm run dev

Or >>> npx create-vite@latest my-app
```

Step 2: Install GitHub Pages package
Open the terminal inside your project and execute:

```
>>> npm install gh-pages --save-dev
```

Step 3: Configure vite.config.js
Open vite.config.js and add the base property.
Suppose your GitHub repository name is **my-app**

```
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'

export default defineConfig({
  plugins: [react()],
  base: '/my-app/',
})
```

Important: The base path must match your GitHub repository name.
For example:

- Repository: portfolio → base: '/portfolio/'
- Repository: student-app → base: '/student-app/'

Step 4: Add deployment scripts in package.json
Open package.json and update the scripts section:

```
"scripts": {
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview",
  "predeploy": "npm run build",
  "deploy": "gh-pages -d dist"
}
```

Keep your other dependencies and configurations unchanged.

Step 5: Create a GitHub repository

1. Open GitHub – Create Repository.
2. Enter repository name: my-app.
3. Select Public (or Private).
4. Click **Create repository**.

Step 6: Push your project to GitHub
Execute these commands from your project directory:

```
>>> git init
>>> git add .
>>> git commit -m "Initial commit"

>>> git branch -M main
>>> git remote add origin https://github.com/USERNAME/my-app.git

>>> git push -u origin main
```

Replace USERNAME with your GitHub username.

Step 7: Deploy your application
Run:

```
>>> npm run deploy
```

This command automatically:

1. Builds your React application.
2. Generates production files inside dist.
3. Publishes the dist folder to a GitHub branch named gh-pages.

Step 8: Enable GitHub Pages
Open your GitHub repository.

1. Go to **Settings**.
2. Select **Pages**.
3. Under Build and deployment, select **Deploy from a branch**.
4. Select branch: gh-pages.
5. Select folder: / (root).
6. Click **Save**.

Your website will be available at: `https://USERNAME.github.io/my-app/`

How to update your website later
Whenever you make changes to your React application, execute:

```
>>> git add .
>>> git commit -m "Updated application"
>>> git push origin main
>>> npm run deploy
```

Common mistake: If your website displays a blank page, check the base property in vite.config.js. It must match your repository name.